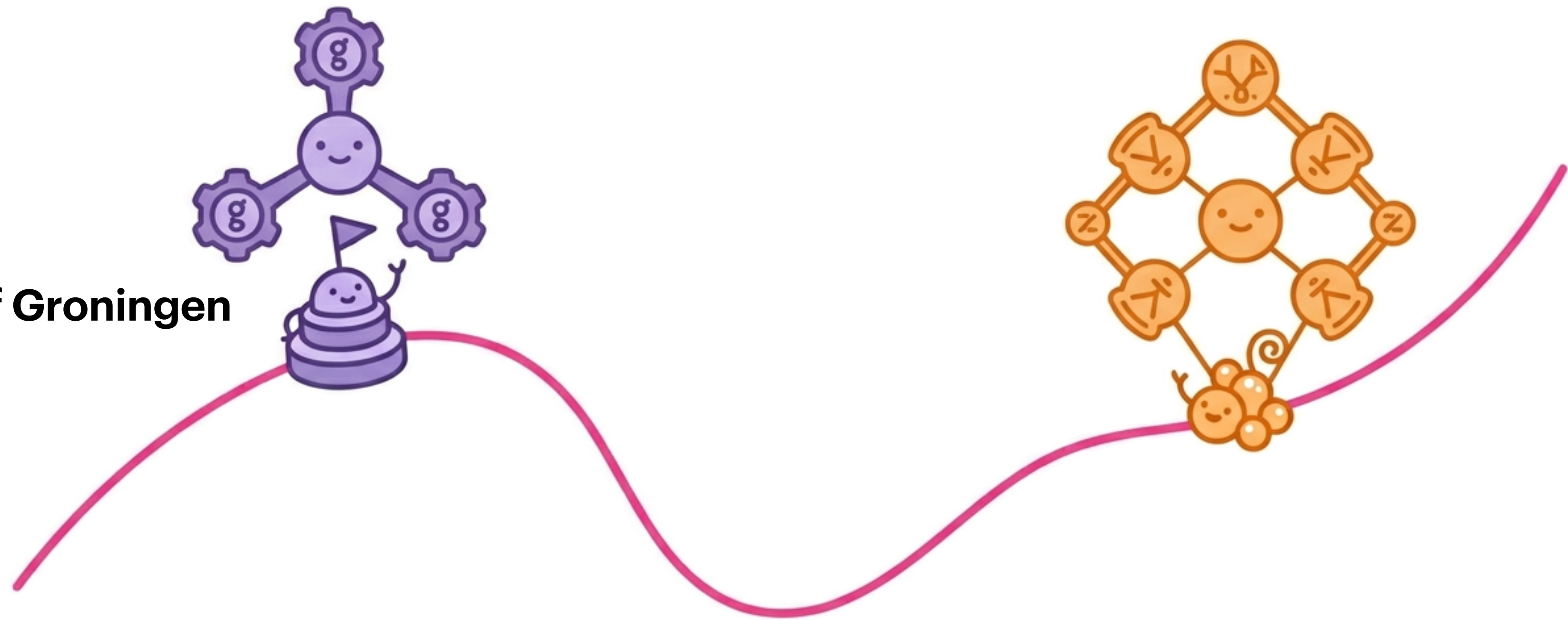


Stacky curves and the local-to-global principle

Juanita Duque-Rosero

Boston University $\longrightarrow$ University of Groningen



Joint work with Christopher Keyes, Andrew Kobin, Manami Roy, Soumya Sankar, and Yidi Wang.

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We can rescale to get infinitely many:

$$x = -3, \quad y = -6, \quad z = -3$$

$$x = 7, \quad y = 14, \quad z = 7$$

$$x = \lambda, \quad y = 2\lambda, \quad z = \lambda$$

Can you find nonzero integers x , y , and z such that

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Definition. A nonzero solution (x, y, z) is **primitive** if $\gcd(x, y, z) = 1$.

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NO: If such a solution existed, then there would be a solution to

$$x^2 + y^2 \equiv 0 \pmod{3}.$$

$y \backslash x$	0	1	2
0	0	1	1
1	1	2	2
2	1	2	2

But then x and y are divisible by 3, which implies that z is too.

Quadratic Forms

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Theorem [Hasse–Minkowski]. Let $f(x_1, \dots, x_n)$ be a non-degenerate quadratic form, where the polynomial f has integer coefficients. Then $f(x_1, \dots, x_n) = 0$ has a non-trivial solution over $\mathbb{Q}$ if and only if it has a nontrivial solution in $\mathbb{Q}_p$ for all primes p and one over $\mathbb{R}$.

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Exercise. Prove that for any $p \geq 7$, there is always a nonzero solution.

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Proof idea. Look at cubes in $(\mathbb{Z}/p\mathbb{Z})^\times$.

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Genus 0 curves

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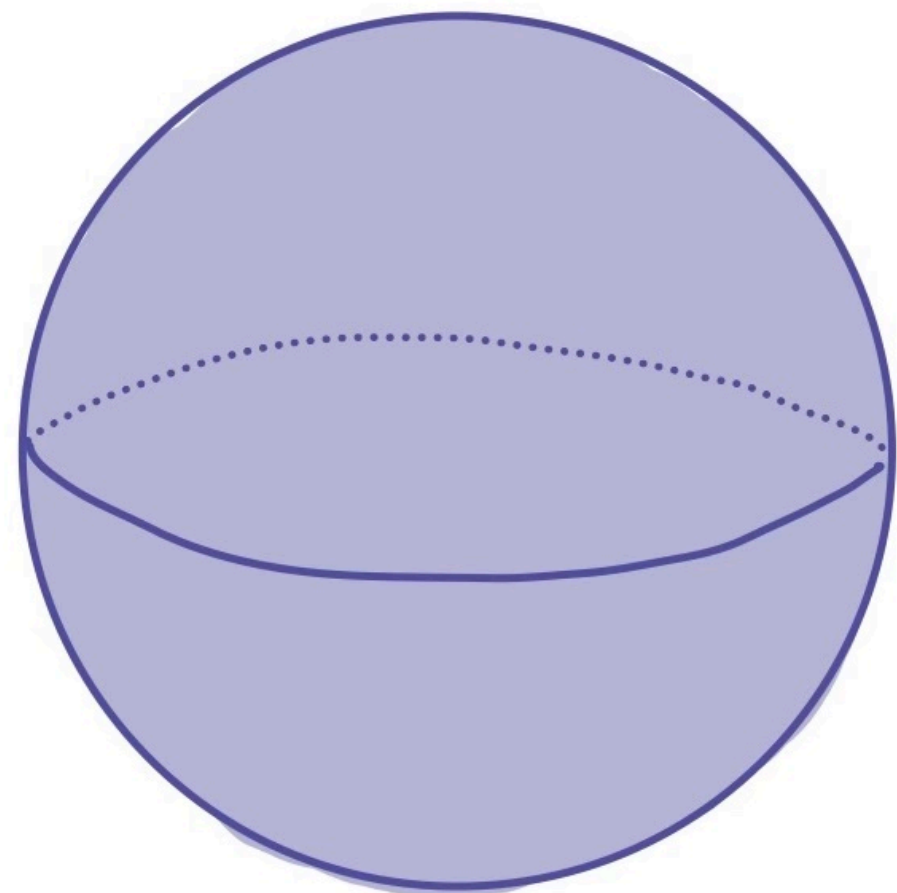
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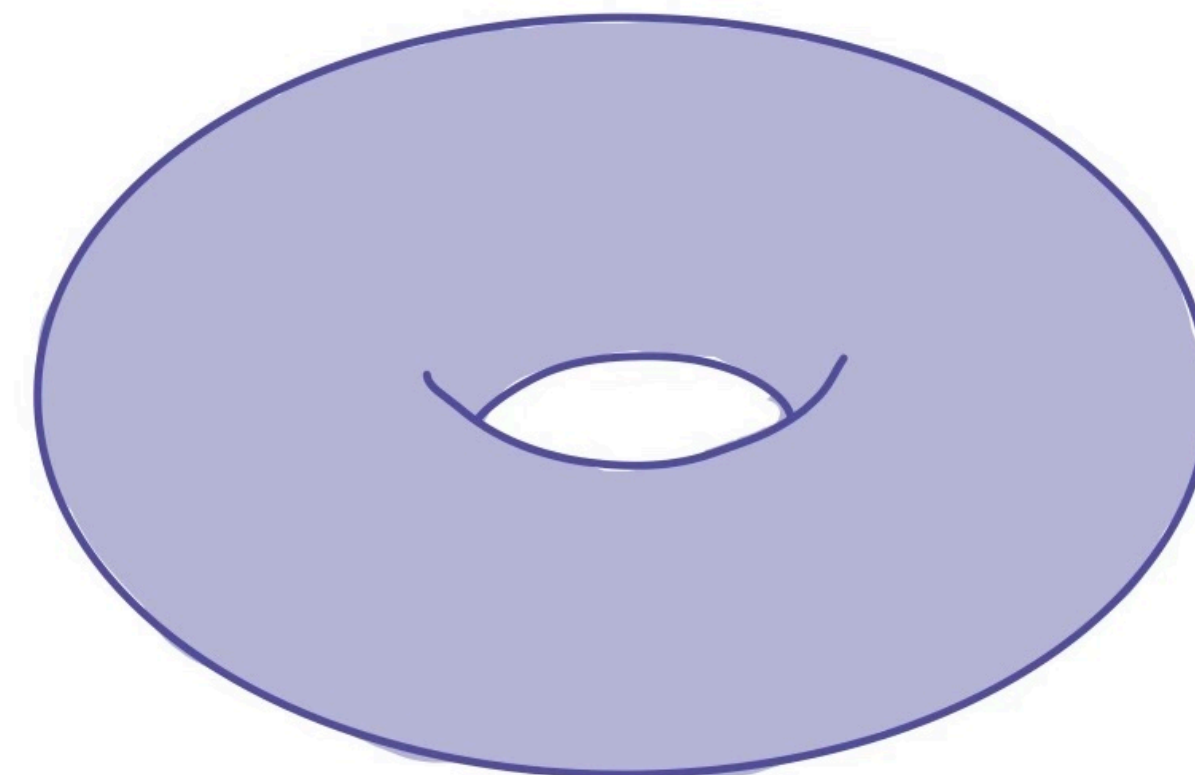
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A solution $(x_0, y_0, z_0) \in \mathbb{Q}^3$ to the equation $f(x, y, z) = 0$ is called a $\mathbb{Q}$ -point on X .

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Note: We can also consider integral points (defined over $\mathbb{Z}$).

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To check if a curve satisfies the local-to-global principle over $\mathbb{Q}$ (similar for $\mathbb{Z}$), we need to check:

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Both answers should be the same.

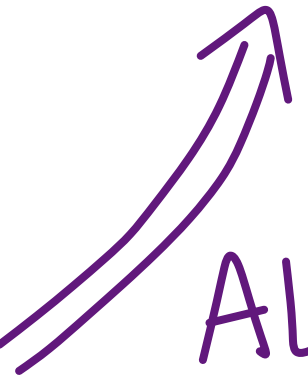
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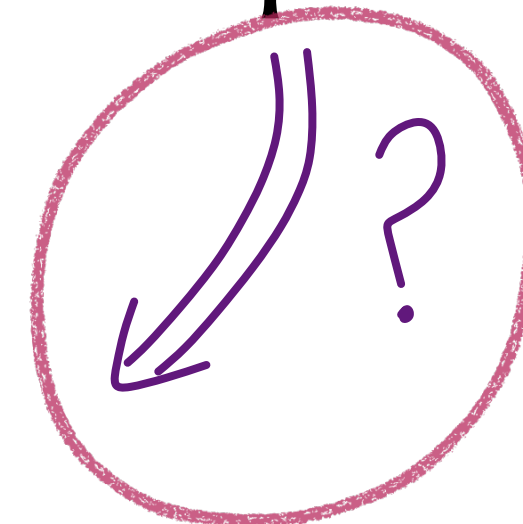
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Theorem [Bhargava '13]. A positive proportion of genus 1 curves in the weighted projective space given by $z^2 = f(x, y)$, where $f(x, y)$ is an integral binary quadratic form, violate the local-to-global principle over $\mathbb{Q}$.

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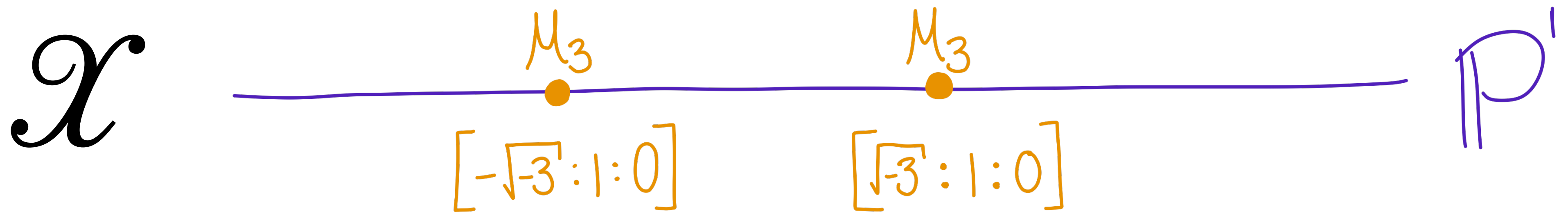
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Primitive integral
solutions to
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Integral points on $\mathcal{X}$:
 $\mathcal{X}(\mathbb{Z})$

Stacky curves

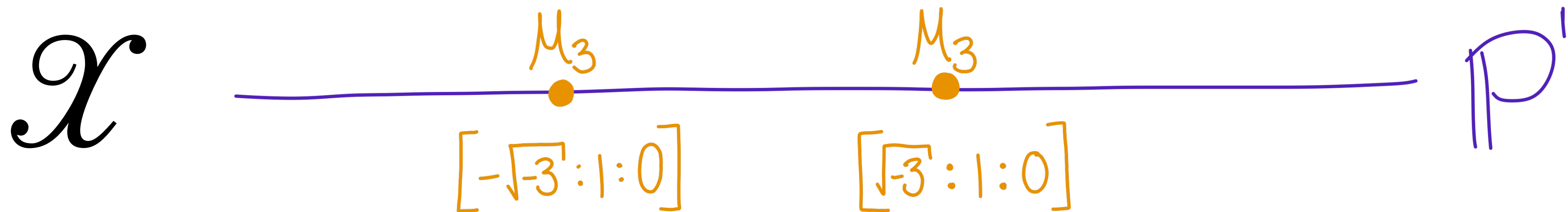
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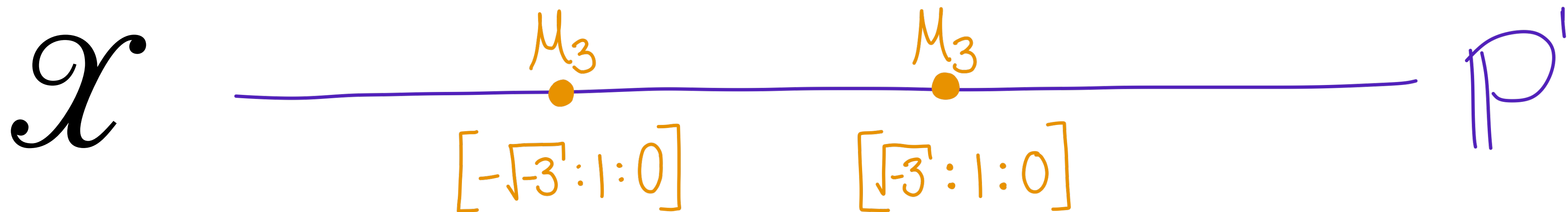
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Claim: $\mathcal{X}$ is a stacky curve of genus $2/3$.

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Theorem [Bhargava & Poonen '21]. Let $\mathcal{X}$ be a stacky curve over $\mathbb{Z}$ of genus less than $1/2$. Then $\mathcal{X}$ satisfies the local-to-global principle.

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Theorem [Bhargava & Poonen '21]. Let
$$S := \operatorname{Spec} \mathbb{Z}[x, y, z] / (z^2 - (3x^2 + xy + 850y^2)) \setminus \{x = y = z = 0\} \subseteq \mathbb{A}_{\mathbb{Z}}^3 \setminus \{(0,0,0)\}$$

Consider the action of μ_2 as $[x : y : z] \mapsto [x : y : \lambda z]$ on $\mathcal{X} := [S/\mathbb{G}_m]$. Let $\mathcal{Y}$ be the quotient stack $[\mathcal{X}/\mu_2]$. Then

- (a) the genus of $\mathcal{Y}$ is $1/2$;
- (b) $\mathcal{Y}(\mathbb{Z}_p) \neq \emptyset$ for every rational prime p and $\mathcal{Y}(\mathbb{R}) \neq \emptyset$;
- (c) $\mathcal{Y}(\mathbb{Z}) \neq \emptyset$.

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Checking for local-to-global

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We need to answer:

1. $\mathcal{X}_{B,C,n}(\mathbb{Z}_p) \neq \emptyset$ for all primes p and $\mathcal{X}_{B,C,n}(\mathbb{R}) \neq \emptyset$?
2. $\mathcal{X}_{B,C,n}(\mathbb{Z}) \neq \emptyset$?

1. Are there local $\mathbb{Z}_p$ -points and $\mathbb{R}$ -points?

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Example $[x^2 + 29y^2 = 3z^3]$.

The ideal 3 is split in $\mathbb{Q}(\sqrt{-29})$. Thus there exist x and y such that

$$x^2 + 29y^2 \equiv 0 \pmod{3}.$$

Conclude $\mathcal{X}_{29,3,3}(\mathbb{Z}_3) \neq \emptyset$.

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- (iii) $C\mathcal{O}_K = \mathfrak{j}\bar{\mathfrak{j}}\mathfrak{r}^2\mathfrak{i}^2$ with $\mathfrak{j}, \mathfrak{r}, \mathfrak{i}$ supported on split, ramified, and inert primes and $\left[\mathfrak{j} \cap \mathbb{Z}[f\sqrt{-B_0}]\right] \in n\text{Cl}\left(\mathbb{Z}[f\sqrt{-B_0}]\right)$.

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Brute force: $(9,0,3), (10,10,10), \dots$, but no primitive solutions.

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Admissible $d \in R_K^\times / (R_K^\times)^3$ satisfy

$$v_{\mathfrak{p}_3}(d) \equiv \pm 1 \pmod{3}, \quad v_{\mathfrak{p}_3}(d) \equiv \mp 1 \pmod{3}, \quad v_{\mathfrak{p}_2}(d) \equiv 0 \pmod{3}.$$

2. Are there global $\mathbb{Z}$ -points?

$$\mathcal{X}_{B,C,n} : x^2 + By^2 = Cz^n \subset \mathbb{P}(n, n, 2)$$

Example $[x^2 + 29y^2 = 3z^3]$.

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Computation of $R_K^\times / (R_K^\times)^3$ reveals no such d exist, so $\mathcal{X}_{29,3,3}(\mathbb{Z}) = \emptyset$.

How many curves $\mathcal{X}_{B,C,3}$ satisfy the local-to-global principle?

$$N^{\text{loc}}(T) := \#\{(B, C) \in \mathbb{Z}^2 : |B|, |C| < T, \mathcal{X}_{B,C,3}(\mathbb{Z}_p) \neq \emptyset \text{ for all primes } p\},$$

$$N(T) := \#\{(B, C) \in \mathbb{Z}^2 : |B|, |C| < T, \mathcal{X}_{B,C,3}(\mathbb{Z}) \neq \emptyset\}.$$

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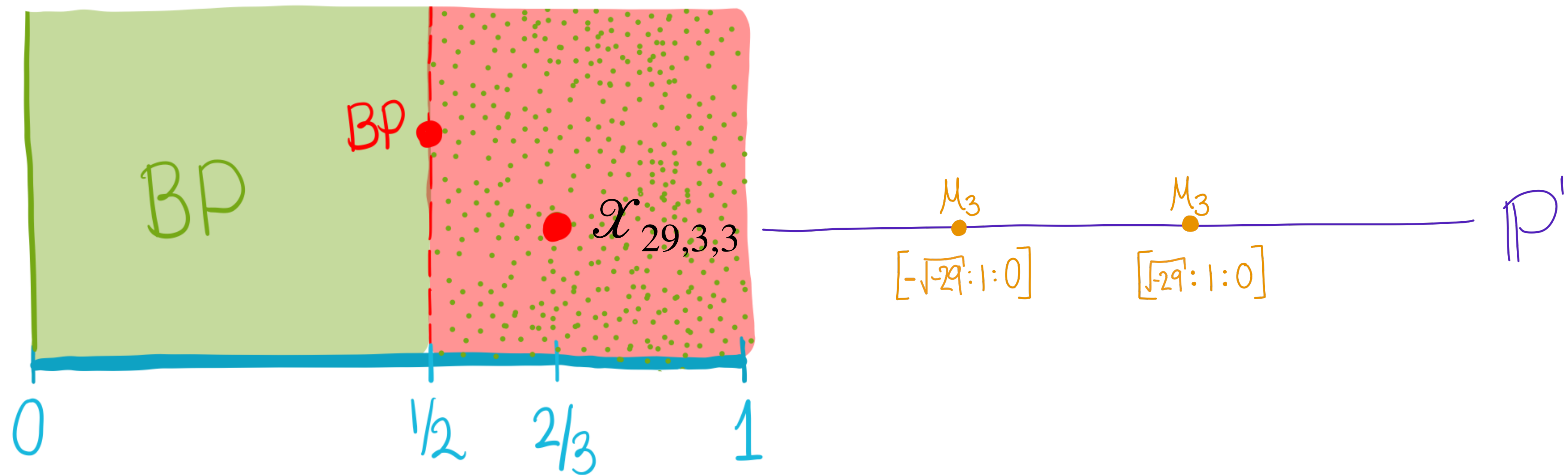
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Theorem [DKKRSW '25]. There is a positive proportion of curves $\mathcal{X}_{B,C,3}$ that non-vacuously satisfy the local-to-global principle for integral points. That is,

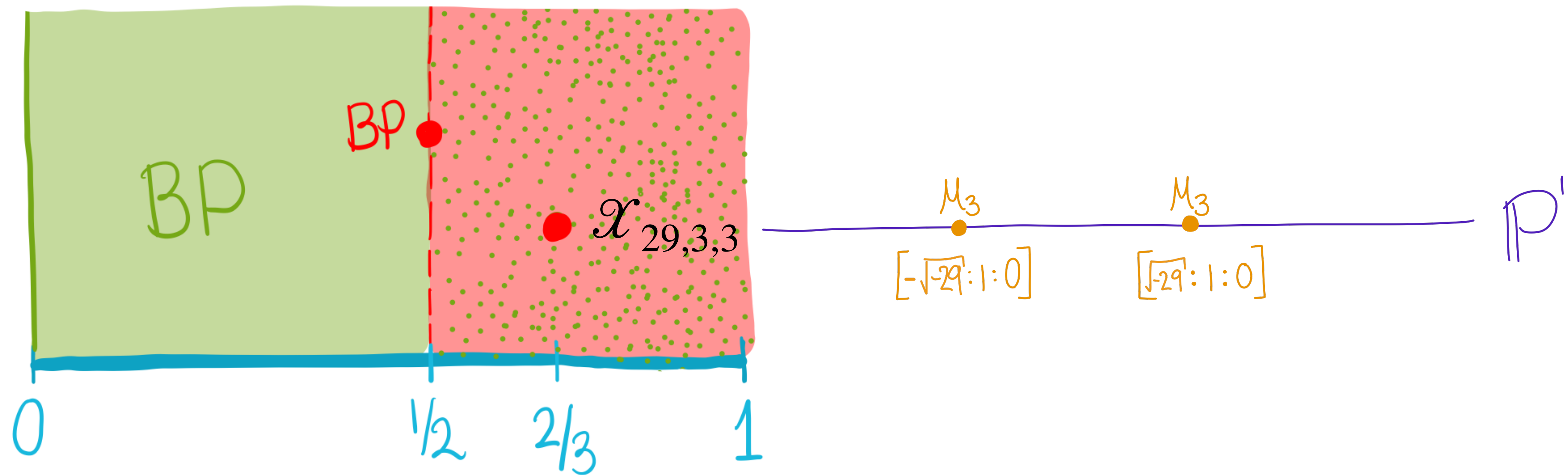
$$\liminf_{T \rightarrow \infty} \frac{N(T)}{N^{\text{loc}}(T)} > 0.$$

Thank you!



Theorem [DKKRSW '25]. There is a positive proportion of curves $\mathcal{X}_{B,C,3}$ that non-vacuously satisfy the local-to-global principle for integral points.

Thank you!



Theorem [DKKRSW '25]. There is a positive proportion of curves $\mathcal{X}_{B,C,3}$ that non-vacuously satisfy the local-to-global principle for integral points.

Q. OK, There are solutions... How many are there? Looking forward to Santi's talk tomorrow!